

Solutions to Homework #1

Problem 1

Let $z = x + jy$, where $x, y \in \mathbb{R}$. We would like show that the following identity holds:

$$\exp(x + jy)^* = \exp(x - jy).$$

The complex conjugate of the left-hand side is written as

$$\exp(x + jy) = \exp(x) \exp(jy) = e^x \cos y + je^x \sin y.$$

Note that e^x , $\cos y$, and $\sin y$ are real-valued. Therefore, we have

$$\exp(x + jy)^* = (e^x \cos y + je^x \sin y)^* = e^x \cos y - je^x \sin y.$$

By Euler's identity, the right-hand side is written as

$$\exp(x - jy) = \exp(x) \exp(-jy) = e^x \cos(-y) + je^x \sin(-y) = e^x \cos y - je^x \sin y.$$

Therefore, we have shown the desired identity.

Problem 2

Note that

$$\underbrace{\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \\ x[4] \end{bmatrix}}_{\mathbf{x}}.$$

It can be rewritten as the following set of equations.

$$\begin{aligned} y[0] &= x[4] \\ y[1] &= x[0] \\ y[2] &= x[1] \\ y[3] &= x[2] \\ y[4] &= x[3]. \end{aligned}$$

By plugging in the entries of $\mathbf{x}$, we have

$$\begin{aligned}\frac{y[0]}{x[0]} &= \frac{x[4]}{x[0]} = \frac{e^{j8\pi/5}}{e^{j0}} = e^{j8\pi/5} \\ \frac{y[1]}{x[1]} &= \frac{x[0]}{x[1]} = \frac{e^{j0}}{e^{j2\pi/5}} = e^{-j2\pi/5} = e^{j8\pi/5} \\ \frac{y[2]}{x[2]} &= \frac{x[1]}{x[2]} = \frac{e^{j2\pi/5}}{e^{j4\pi/5}} = e^{-j2\pi/5} = e^{j8\pi/5} \\ \frac{y[3]}{x[3]} &= \frac{x[2]}{x[3]} = \frac{e^{j4\pi/5}}{e^{j6\pi/5}} = e^{-j2\pi/5} = e^{j8\pi/5} \\ \frac{y[4]}{x[4]} &= \frac{x[3]}{x[4]} = \frac{e^{j6\pi/5}}{e^{j8\pi/5}} = e^{-j2\pi/5} = e^{j8\pi/5}.\end{aligned}$$

This set of equations can be rewritten as

$$\underbrace{\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix}}_{\mathbf{y}} = e^{j8\pi/5} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \\ x[4] \end{bmatrix}}_{\mathbf{x}}.$$

Therefore, it follows that $\mathbf{x}$ is an eigenvector of $\mathbf{A}$, where the corresponding eigenvalue is $\lambda = e^{j8\pi/5}$.

Problem 3

Note that the output $y[n]$ is obtained by stretching the input $x[n]$ along the horizontal (time) axis with respect to the origin ($n = 0$).

Linearity: Let $x_3[n] = ax_1[n] + bx_2[n]$. In one hand, for any even n , we have

$$H\{x_3[n]\} = x_3[n/2] = ax_1[n/2] + bx_2[n/2] = aH\{x_1[n]\} + bH\{x_2[n]\}.$$

On the other hand, for any odd n , we have

$$H\{x_3[n]\} = 0 = a \cdot 0 + b \cdot 0 = aH\{x_1[n]\} + bH\{x_2[n]\}.$$

Therefore, for all $n \in \mathbb{Z}$, we have

$$H\{x_3[n]\} = aH\{x_1[n]\} + bH\{x_2[n]\}.$$

We did not assume any condition on a , b , $x_1[n]$, and $x_2[n]$. In other words, the identity holds in general. Hence the system is linear.

Time-invariance: Note that for any input $x[n]$, the corresponding output $y[n]$ satisfies $y[n] = 0$ for any odd time index n . Therefore, $z[n] = y[n - 1]$ satisfies that $z[n] = 0$ for any even time index n . However, $w[n] = H\{x[n - 1]\}$ satisfies $w[n] = 0$ for any odd time index n . Therefore $w[n] = z[n]$ cannot hold in general. Therefore, the system is time-variant.

Alternatively, we may consider a simple counter example. Let $x[n] = \delta[n]$. Then it follows that $y[n] = \delta[n]$. However,

$$H\{x[n - 1]\} = H\{\delta[n - 1]\} = \delta[n - 2]$$

is different from $y[n - 1] = \delta[n - 1]$.

Causality: Note that $y[-2] = x[-1]$. For this instance, the output value refers to a future value of the input. Hence, the system is not causal.

Stability: We consider the largest magnitude of $y[n]$, which satisfies

$$\max_{n \in \mathbb{Z}} |y[n]| = \max_{m \in \mathbb{Z}} |y[2m]|.$$

This holds because for all odd indices the output takes zero. Moreover by the system definition we have

$$\max_{n \in \mathbb{Z}} |y[n]| = \max_{m \in \mathbb{Z}} |y[2m]| = \max_{m \in \mathbb{Z}} |x[m]|.$$

Note that the last term is the largest magnitude of the input. Then we deduce that the largest magnitudes of the input and output always coincide. Therefore, for any bounded input, the output signal is also bounded. Then we conclude that the system is stable.

Problem 4

We first recognize that the output is given as the convolution of the input and $h(t)$, that is,

$$y(t) = h(t) * x(t).$$

We would like to use various properties of the convolution operator.

- (a) When the input $x(t)$ is $\delta(t)$, by the definition of the system, the output $y(t)$ is given by

$$y(t) = h(t) * \delta(t) = h(t).$$

This follows from the fact that $\delta(t)$ is the unity for the convolution operator. We therefore conclude that $h(t)$ indeed turns out the impulse response of the given convolutive system.

- (b) For simplicity, we will use the system action notation $H\{\cdot\}$. Recall that $H\{x(t)\}$ denotes the output signal from the input $x(t)$ through the system H . To show the linearity of the system, it suffices to show

$$H\{ax_1(t) + bx_2(t)\} = aH\{x_1(t)\} + bH\{x_2(t)\} \quad (1)$$

for arbitrary signals $x_1(t)$, $x_2(t)$ and for arbitrary scalars a, b .

The left-hand side of (1) is computed as follows:

$$\begin{aligned}
H\{ax_1(t) + bx_2(t)\} &= h(t) * (ax_1(t) + bx_2(t)) && \because \text{definition of } H \\
&= h(t) * (ax_1(t)) + h(t) * (bx_2(t)) && \because \text{distributivity of convolution} \\
&= a(h(t) * x_1(t)) + b(h(t) * x_2(t)) && \because \text{homogeneity of convolution} \\
&= aH\{x_1(t)\} + bH\{x_2(t)\}, && \because \text{definition of } H
\end{aligned}$$

where the last expression coincides with the right-hand side. Here the homogeneity of the convolution follows from the definition, i.e.,

$$h(t) * ax(t) = \int_{-\infty}^{\infty} h(\lambda)ax(t - \lambda)d\lambda = a \left(\int_{-\infty}^{\infty} h(\lambda)x(t - \lambda)d\lambda \right) = a(h(t) * x(t)).$$

Since signals $x_1(t)$, $x_2(t)$ and scalars a, b were arbitrary, we have shown that (1) holds in general. Therefore the system is linear.

(c) To show that the system is time-invariant, it suffices to show

$$H\{\delta(t - d) * x_1(t)\} = \delta(t - d) * H\{x_1(t)\} \quad (2)$$

for arbitrary signals $x_1(t)$ and for arbitrary delay d . The left-hand side of (2) is computed as follows:

$$\begin{aligned}
H\{\delta(t - d) * x_1(t)\} &= h(t) * (\delta(t - d) * x_1(t)) && \because \text{definition of } H \\
&= (h(t) * \delta(t - d)) * x_1(t) && \because \text{associativity of convolution} \\
&= (\delta(t - d) * h(t)) * x_1(t) && \because \text{commutativity of convolution} \\
&= \delta(t - d) * (h(t) * x_1(t)) && \because \text{associativity of convolution} \\
&= \delta(t - d) * H\{x_1(t)\}, && \because \text{definition of } H
\end{aligned}$$

where the last expression coincides with the right-hand side. Since signals $x_1(t)$ and delay d were arbitrary, we have shown (2) in general. Therefore the system is time-invariant.

Problem 5

(a) Let $z[n] = ax[n] + by[n]$. Then we have

$$\mathcal{D}\{ax[n] + by[n]\} = \mathcal{D}\{z[n]\} = z[n - 1] = ax[n - 1] + by[n - 1] = a\mathcal{D}\{x[n]\} + b\mathcal{D}\{y[n]\}.$$

Since $x[n]$, $y[n]$, a , and b were arbitrary, we conclude that the delay operator $\mathcal{D}$ is linear.

(b) Let $z[n] = ax[n] + by[n]$. Then we have

$$\begin{aligned}
\mathcal{V}\{ax[n] + by[n]\} &= \mathcal{V}\{z[n]\} \\
&= z[n] - z[n-1] \\
&= ax[n] + by[n] - ax[n-1] - by[n-1] \\
&= ax[n] - ax[n-1] + by[n] - by[n-1] \\
&= a\mathcal{V}\{x[n]\} + b\mathcal{V}\{y[n]\}.
\end{aligned}$$

Since $x[n]$, $y[n]$, a , and b were arbitrary, we conclude that the differentiation operator $\mathcal{V}$ is linear.

(c) Let $x[n] = \delta[n]$ and $a = -1$. Then

$$\mathcal{S}\{ax[n]\} = \mathcal{S}\{-\delta[n]\} = \delta[n]$$

whereas

$$a\mathcal{S}\{x[n]\} = -\delta[n].$$

This counter example shows that the squaring operator is not homogeneous. Recall that the linearity implies both homogeneity and additivity. Since the homogeneity is not satisfied, the system is not linear. Note that there are other counter examples.

(d) The matrix corresponding to the difference operator is given by

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

(e) The matrix multiplication yields

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_0 + x_1 \\ x_0 + x_1 + x_2 \\ x_0 + x_1 + x_2 + x_3 \end{bmatrix},$$

which corresponds to the cumulative sum. This operation is the discrete-time analog of anti-derivative.